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CE587 Homework 1

Structural Optimization

Fall Semester 2026–2027

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GitHub Repository: `https://github.com/ImranMETU/CE587_Assignment1`

1 Question 1 — Hollow circular beam-column

1.1 Mathematical formulation (part a)

A uniform thin-walled circular member of length $L = 3$ m is designed for an axial load $P = 50$ kN and self-weight. The axial-stress case and the self-weight deflection case are evaluated separately, as prescribed. The material data are $E = 210$ GPa, $\rho = 7800$ kg/m^3, allowable stress $\sigma_a = 250$ MPa and $g = 9.80$ m/s^2. The design bounds are $0.02 \le R \le 0.20$ m and $0.001 \le t \le 0.010$ m.

Design variables. $\mathbf{x}_1 = [R, t]^T$.

Using the thin-wall expressions supplied for the model,

$$A = 2\pi Rt, \qquad I = \pi R^3 t, \qquad w = \rho g A, \tag{1}$$

$$\sigma = \frac{P}{A}, \qquad \min_{R,t}\ m(R,t) = \rho L A = 2\pi\rho L R t. \tag{2}$$

The assignment gives $\delta = 5wL^4/(384EI)$. Substituting A, I and w yields

$$\delta = \frac{5\rho g(2\pi Rt)L^4}{384E\pi R^3 t} = \frac{5\rho g L^4}{192ER^2}. \tag{3}$$

The thickness cancels because self-weight and flexural stiffness both vary linearly with t. With $\delta_a = 0.001L = 3$ mm, the normalized constraints $g_i \le 0$ are

$$g_1 = \frac{P}{2\pi Rt\sigma_a} - 1, \qquad g_2 = \frac{5\rho g L^4}{192ER^2\delta_a} - 1, \tag{4}$$

$$g_3 = \frac{R}{20t} - 1, \qquad g_4 = \frac{0.02}{R} - 1, \tag{5}$$

$$g_5 = \frac{R}{0.20} - 1, \qquad g_6 = \frac{0.001}{t} - 1, \tag{6}$$

$$g_7 = \frac{t}{0.010} - 1. \tag{7}$$

Objective function. $\min_{\mathbf{x}_1} f_1(\mathbf{x}_1) = \rho L A = 2\pi\rho L R t$.

Constraints. The normalized inequalities are $g_i(\mathbf{x}_1) \le 0$ for $i = 1, \dots, 7$, as defined in (7). The lower bounds keep area and inertia positive. No additional stability or manufacturing limit is imposed.

1.2 Graphical solution (part b)

For graphing, use centimetres. The axial-stress boundary is $Rt = 1/\pi$ cm^2, or $t = 1/(\pi R)$, and the radius-to-thickness boundary is $t = R/20$. The deflection requirement gives $R \ge 1.5998$ cm. Since the assigned lower radius is 2 cm, deflection is redundant; at $R = 2$ cm, $\delta = 1.920$ mm, below 3 mm. The feasible region lies above

$$t_c(R) = \max\left\{0.1, \frac{1}{\pi R}, \frac{R}{20}\right\}, \qquad 2 \le R \le 20 \text{ cm}, \tag{8}$$

and below $t = 1$ cm. The dashed curve is the composite constraint surface (CCS). Figure 1 focuses on the feasible corner so the full optimal segment and its representative endpoint can be distinguished. The 4.68, 5, 6 and 8 kg contours are shown.

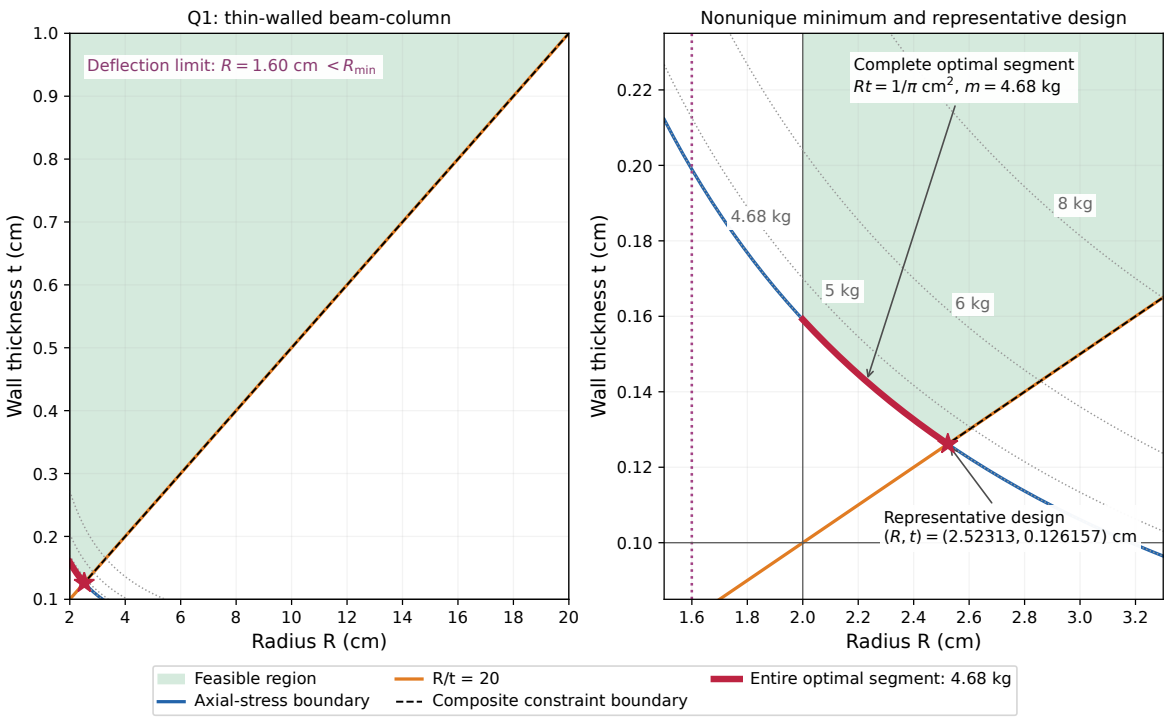


Figure 1: Q1 full design domain (left, $2 \le R \le 20$ cm and $0.1 \le t \le 1$ cm) and close-up of the optimum (right). Both panels show the feasible region, constraint boundaries, CCS and mass contours; the red curve is the complete minimum-mass set and the star marks its representative endpoint.

Since axial stress requires $A \ge P/\sigma_a$, every feasible design satisfies

$$m = \rho L A \ge \rho L \frac{P}{\sigma_a} = 4.680 \text{ kg}. \tag{9}$$

Equality holds when $Rt = 1/\pi$ cm^2. Intersecting this curve with $R/t \le 20$ and the radius and thickness bounds gives the complete optimal set

$$\boxed{2.000 \le R \le \sqrt{20/\pi} = 2.52313 \text{ cm}, \qquad t = \frac{1}{\pi R} \text{ cm}.} \tag{10}$$

Thus the minimum is nonunique. The requested representative is the right endpoint, $(R, t) = (2.52313, 0.126157)$ cm, where axial stress and R/t are active.

1.3 Microsoft Excel Solver (part c)

The workbook sheet `Q1_BeamColumn` uses centimetres for the changing cells and converts to SI in its response formulas. It calculates area, inertia, distributed self-weight, stress, deflection, mass and normalized constraint residuals. The saved Solver model uses the settings in Table 1.

Table 1: Q1 Solver configuration and design checks.

Solver setting	Cell reference	Configuration / value	Result at representative design
Objective	B4	Minimize	$m = 4.6800$ kg
Changing cells	B7:B8	Radius and thickness (cm)	$R = 2.52313$, $t = 0.126157$ cm
Constraints	B29:B35 $\leq$ C29:C35	Normalized constraints	See checks below
Explicit bounds	G11:G14; B7:B8	$2 \leq R \leq 20$, $0.1 \leq t \leq 1$ cm	Bounds inactive
Method	–	GRG Nonlinear	Saved native Solver model
Precision / convergence	–	$10^{-9}/10^{-9}$	–
Iterations / time	–	1500 / 100 s	–
Scaling / derivatives	–	Automatic / central	–
Initial design	B7:B8	$R = 3$ cm, $t = 0.15$ cm	–
Axial stress	B29	$\sigma/\sigma_a \leq 1$	250.000 MPa / 250 MPa, active
Self-weight deflection	B30	$\delta/\delta_a \leq 1$	1.206 mm / 3 mm, inactive
R/t limit	B31	$(R/t)/20 \leq 1$	20.000 / 20, active
Radius / thickness bounds	B32:B35	Lower and upper limits	Inactive

Excel returned $R = 2.52313$ cm, $t = 0.126157$ cm and $m = 4.68000$ kg. The returned stress is within the configured Solver tolerance; the analytical point satisfies the limit exactly. Python gives the same dimensions and mass to the shown precision. Appendix A presents the worksheet and actual Solver Parameters dialog for Question 1; Table 1 summarizes the settings.

2 Question 2 — Hollow circular cantilever

2.1 Mathematical formulation (part d)

A cantilever of length $L = 10$ m carries a tip force $P = 14$ kN. The material density is 7850 kg/m^3; the allowable bending and shear stresses are 165 MPa and 50 MPa.

Design variables. $\mathbf{x}_2 = [R_o, R_i]^T$, with $0 \le R_i < R_o \le 0.40$ m.

For an exact annular section,

$$A = \pi(R_o^2 - R_i^2), \qquad I = \frac{\pi}{4}(R_o^4 - R_i^4), \tag{11}$$

$$\sigma = \frac{PLR_o}{I}, \qquad \tau = \frac{P(R_o^2 + R_oR_i + R_i^2)}{3I}, \tag{12}$$

$$\min \quad m(R_o, R_i) = \rho LA. \tag{13}$$

Objective function. $\min_{\mathbf{x}_2} f_2(\mathbf{x}_2) = \rho LA$ as given in (13).

Constraints. With $r_{\max} = 0.40$ m, the normalized stress and radius inequalities are $g_i(\mathbf{x}_2) \le 0$ for $i = 1, \ldots, 5$; the strict geometric condition is $g_6(\mathbf{x}_2) < 0$:

$$g_1 = \sigma/(165 \text{ MPa}) - 1, \qquad g_2 = \tau/(50 \text{ MPa}) - 1, \tag{14}$$

$$g_3 = R_o/r_{\max} - 1, \qquad g_4 = R_i/r_{\max} - 1, \tag{15}$$

$$g_5 = -R_i/r_{\max}, \qquad g_6 = (R_i - R_o)/r_{\max} < 0. \tag{16}$$

The last relation is strict in the mathematical model. The workbook uses a 10^{-8} m positive-thickness guard to keep Solver away from the singular zero-area section; it is a numerical safeguard, not a manufacturing requirement.

2.2 Graphical solution (part e)

At a fixed R_o, increasing R_i reduces the area and mass, so the upper feasible boundary is set by the first bending or shear limit reached. The bending boundary is

$$R_{i,b}(R_o) = \left(R_o^4 - CR_o\right)^{1/4}, \qquad C = \frac{4PL}{\pi\sigma_b}, \qquad R_o \ge C^{1/3}, \tag{17}$$

and the shear boundary follows from $\tau(R_o, R_i) = \tau_a$. Figure 2 shows the feasible region, both stress boundaries, the excluded equal-radii line, the CCS and mass contours. Its close view places the outer-radius limit and the optimum at $R_o = 40$ cm on the same edge.

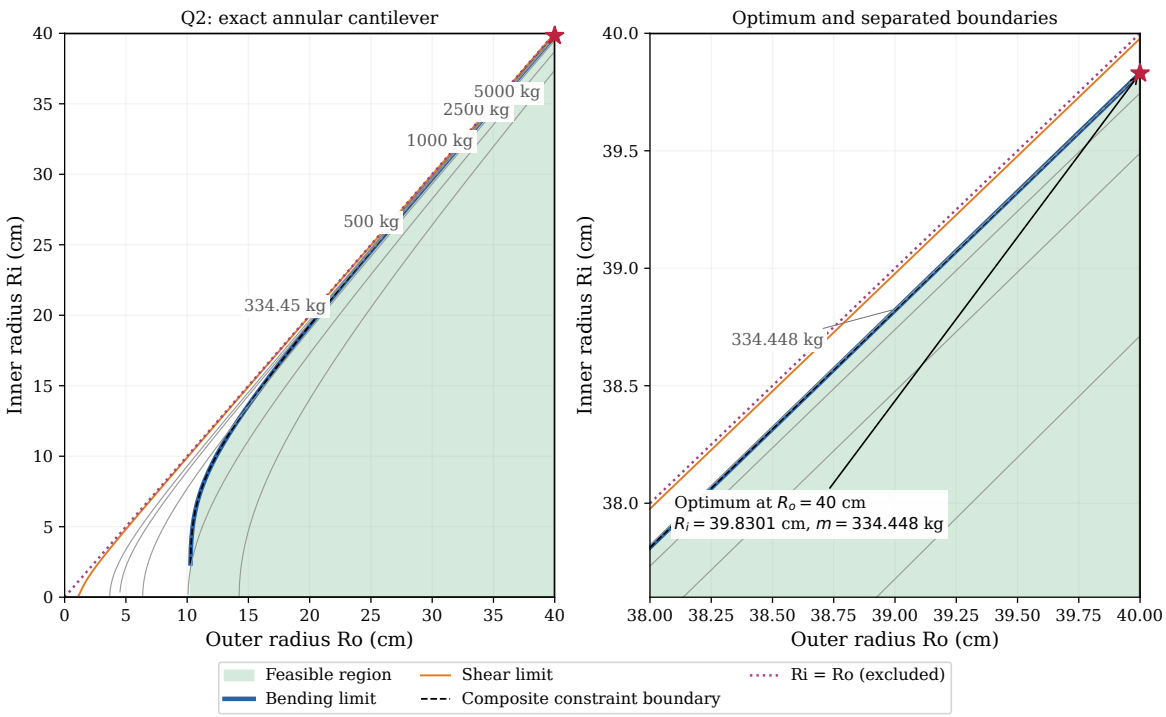


Figure 2: Q2 full design domain (left, $0 \le R_o, R_i \le 40$ cm) and close-up of the optimum (right). The bending and shear boundaries, excluded equal-radii line, CCS, mass contours and radius bound are shown in both panels. The star marks the optimum at $R_o = 40$ cm.

Shear is redundant for every bending-feasible section because

$$\frac{\tau}{\sigma} = \frac{R_o^2 + R_o R_i + R_i^2}{3LR_o} \le \frac{R_o}{L} \le 0.04, \qquad \tau \le 6.6 \text{ MPa} < 50 \text{ MPa}. \tag{18}$$

On the bending boundary the area can be written

$$A_b(R_o) = \frac{\pi C}{R_o + \sqrt{R_o^2 - C/R_o}}. \tag{19}$$

The denominator increases with R_o, so the minimum mass decreases up to the assigned upper limit. The global optimum is therefore

$$\boxed{R_o^* = 0.400000 \text{ m}, \quad R_i^* = 0.398301 \text{ m}, \quad m^* = 334.448 \text{ kg}.} \tag{20}$$

Bending and the outer-radius bound are active; shear is inactive. The wall is 1.69880 mm thick. This mathematical minimum would need local-buckling and fabrication checks in an actual design; those requirements are outside the assigned optimization model.

2.3 Microsoft Excel Solver (part f)

The `Q2_Cantilever` sheet contains the annular properties, stresses, mass and constraint utilizations. Solver minimizes B4 by changing B7:B8. The stress rows use algebraically equivalent polynomial residuals, so trial evaluations remain finite near the geometric boundary; the reported design is checked against the original stress equations.

Table 2: Q2 Solver configuration and final checks.

Solver setting	Cell reference	Configuration / value	Result at optimum
Objective	B4	Minimize	$m = 334.448$ kg
Changing cells	B7:B8	Outer and inner radii (cm)	$R_o = 40.0000$, $R_i = 39.8301$ cm
Stress constraints	B45:B46 ≤ 0	Smooth polynomial bending/shear residuals	Checked against stress equations
Radius constraints	B28:B31 $\leq$ C28:C31; B7:B8	Radius limits 0–40 cm	$R_o = 40$ cm, active
Geometry guard	B22 $\geq$ B16	$\epsilon = 10^{-8}$ m	$R_o - R_i = 1.69880$ mm
Method	–	GRG Nonlinear	Saved native Solver model
Precision / convergence	–	$10^{-9}/10^{-9}$	–
Iterations / time	–	2000 / 100 s	–
Scaling / derivatives	–	Automatic / central	–
Initial design	B7:B8	$R_o = 30$ cm, $R_i = 20$ cm	–
Bending stress	B26	$\sigma/\sigma_b \leq 1$	165.000 MPa / 165 MPa, active
Shear stress	B27	$\tau/\tau_a \leq 1$	6.572 MPa / 50 MPa, inactive
Mass	B4	kg	334.448 kg

Excel and Python agree within the saved Solver tolerance. Appendix A presents the worksheet and actual Solver Parameters dialog for Question 2; Table 2 summarizes the settings.

3 Question 3 — Minimum-mass tripod

3.1 Mathematical formulation (part a)

Three identical solid circular struts support an apex load $W = 60$ kN over an equilateral base with side $B = 120$ cm. The material is aluminum alloy 2014-T6 with $E = 75$ GPa, $\rho = 2800$ kg/m^3 and allowable compression stress $\sigma_a = 150$ MPa. The safety factor is $FS = 2$. Calculations use N and cm: $E = 7.5 \times 10^6$ N/cm^2, $\rho = 0.0028$ kg/cm^3, and $\sigma_a = 15000$ N/cm^2.

Design variables. $\mathbf{x}_3 = [H, D]^T$, with $50 \le H \le 200$ cm and $0.5 \le D \le 10$ cm.

Each base vertex is a horizontal distance $B/\sqrt{3}$ from the centroid, so the strut length is $l = \sqrt{H^2 + B^2/3}$. Symmetry cancels horizontal force components, while vertical equilibrium gives $3P(H/l) = W$. Thus

$$P = \frac{Wl}{3H}, \quad A = \frac{\pi D^2}{4}, \quad I = \frac{\pi D^4}{64}, \quad \sigma = \frac{P}{A}, \quad P_{cr} = \frac{\pi^2 EI}{l^2}, \quad \min\ m(H, D) = 3\rho Al. \tag{21}$$

Objective function. $\min_{\mathbf{x}_3} f_3(\mathbf{x}_3) = 3\rho Al$ as given in (21).

Constraints. The normalized inequalities are $g_i(\mathbf{x}_3) \le 0$ for $i = 1, \ldots, 6$:

$$g_1 = \frac{P}{A\sigma_a} - 1, \qquad g_2 = \frac{FS\,P}{P_{cr}} - 1, \tag{22}$$

$$g_3 = \frac{50}{H} - 1, \qquad g_4 = \frac{H}{200} - 1, \tag{23}$$

$$g_5 = \frac{0.5}{D} - 1, \qquad g_6 = \frac{D}{10} - 1. \tag{24}$$

The prescribed strut length is used in Euler's formula and the safety factor is applied once.

3.2 Graphical solution (part b)

Solving the stress and buckling conditions for diameter gives

$$D_s(H) = \left(\frac{4Wl}{3\pi\sigma_a H}\right)^{1/2}, \qquad D_b(H) = \left(\frac{64FS\,Wl^3}{3\pi^3 EH}\right)^{1/4}, \qquad D_c(H) = \max\{0.5, D_s, D_b\}. \tag{25}$$

Figure 3 separates the compression-strength boundary from the Euler buckling boundary and shows their composite boundary with mass contours. In the assigned height range $D_b > D_s$; on the buckling boundary the maximum compression utilization is 0.247553 at $H = 50$ cm. Thus buckling governs, while the compressive-strength constraint is inactive throughout.

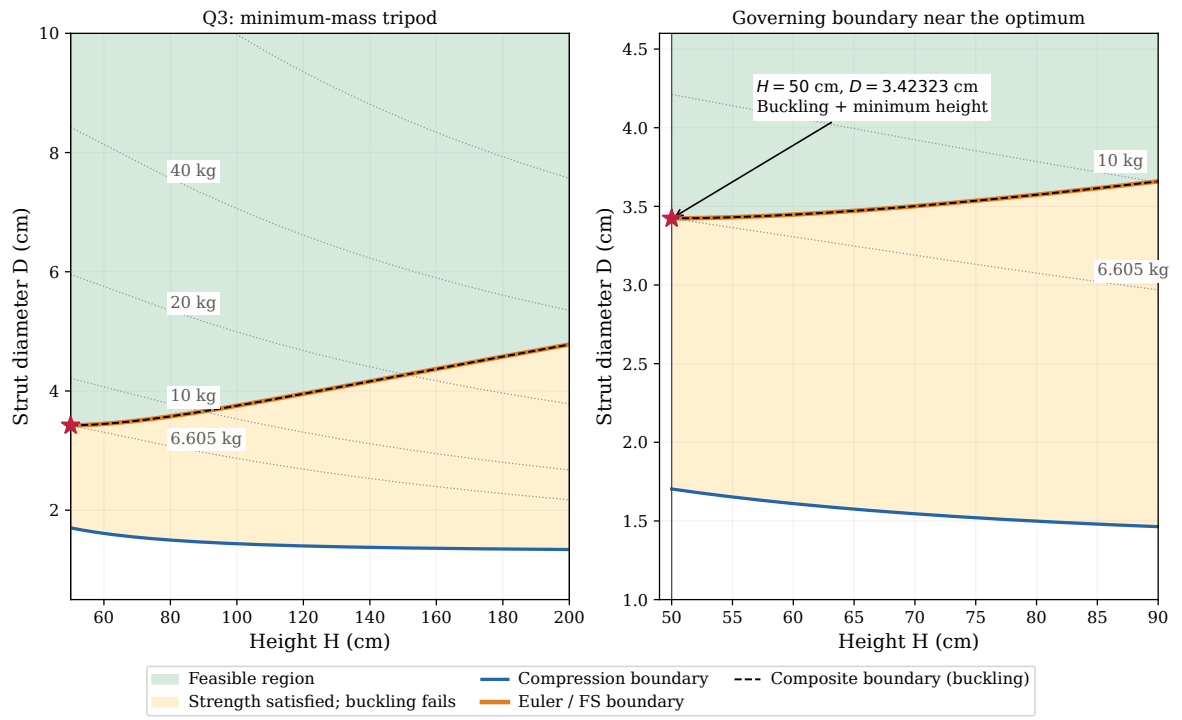


Figure 3: Q3 full design domain (left, $50 \le H \le 200$ cm and $0.5 \le D \le 10$ cm) and close-up near the minimum height (right). The compression-strength and Euler boundaries are distinct; the dashed CCS follows the buckling boundary. Amber indicates designs that pass strength but fail buckling. The star marks the optimum.

For fixed H, mass increases with D, so the lightest feasible section lies at $D = D_b(H)$. Along that boundary $m_b(H)$ is proportional to $l^{5/2}H^{-1/2}$, and

$$\frac{d\log m_b}{dH} = \frac{4H^2 - B^2/3}{2H(H^2 + B^2/3)} > 0 \qquad (H \ge 50 \text{ cm}). \tag{26}$$

The unconstrained stationary height $B/\sqrt{12} = 34.641$ cm is below the permitted range. Therefore the minimum occurs at the lower height bound, with buckling active:

$$\boxed{H^* = 50.0000 \text{ cm}, \quad D^* = 3.42323 \text{ cm}, \quad m^* = 6.60545 \text{ kg}.} \tag{27}$$

3.3 Microsoft Excel Solver (part c)

The `Q3_Tripod` worksheet calculates strut geometry, equilibrium force, section properties, compression, Euler load, total mass and all six constraint utilizations. Solver minimizes B4 by changing B7:B8.

Table 3: Q3 Solver configuration and final checks.

Solver setting	Cell reference	Configuration / value	Result at optimum
Objective	B4	Minimize	$m = 6.60545$ kg
Changing cells	B7:B8	Height and diameter (cm)	$H = 50.0000$, $D = 3.42323$ cm
Constraints	B26:B31 $\leq$ C26:C31	All six normalized constraints	–
Method	–	GRG Nonlinear	Saved native Solver model
Precision / convergence	–	$10^{-9}/10^{-9}$	–
Iterations / time	–	2000 / 100 s	–
Scaling / derivatives	–	Automatic / central	–
Initial design	B7:B8	$H = 100$ cm, $D = 5$ cm	–
Compression stress	B26	$\sigma/\sigma_a \leq 1$	37.133 MPa / 150 MPa, inactive
Euler buckling	B27	$FS\,P/P_{cr} \leq 1$	$P_{cr} = 68.352$ kN $= 2P$, active
Height lower bound	B28	$H \geq 50$ cm	Active
Mass	B4	kg	6.60545 kg

Excel and Python agree within the saved Solver tolerance. Appendix A presents the worksheet and actual Solver Parameters dialog for Question 3; Table 3 summarizes the settings.

4 Final results and discussion

Table 4 collects the representative designs and their governing constraints. Excel may return any feasible point on the Q1 optimal segment, with a different radius and thickness but the same minimum mass of 4.68 kg. Q2 is controlled by bending at the outer-radius limit, and Q3 by Euler buckling at the minimum height. The mass difference is $\Delta m = |m_{\text{Excel}} - m_{\text{Python}}|$, calculated before rounding the displayed results. The differences are consistent with numerical tolerances. Fresh Excel Solver runs from different starting designs reproduced all three saved masses; Appendix B gives the starting designs, results, and Solver confirmations.

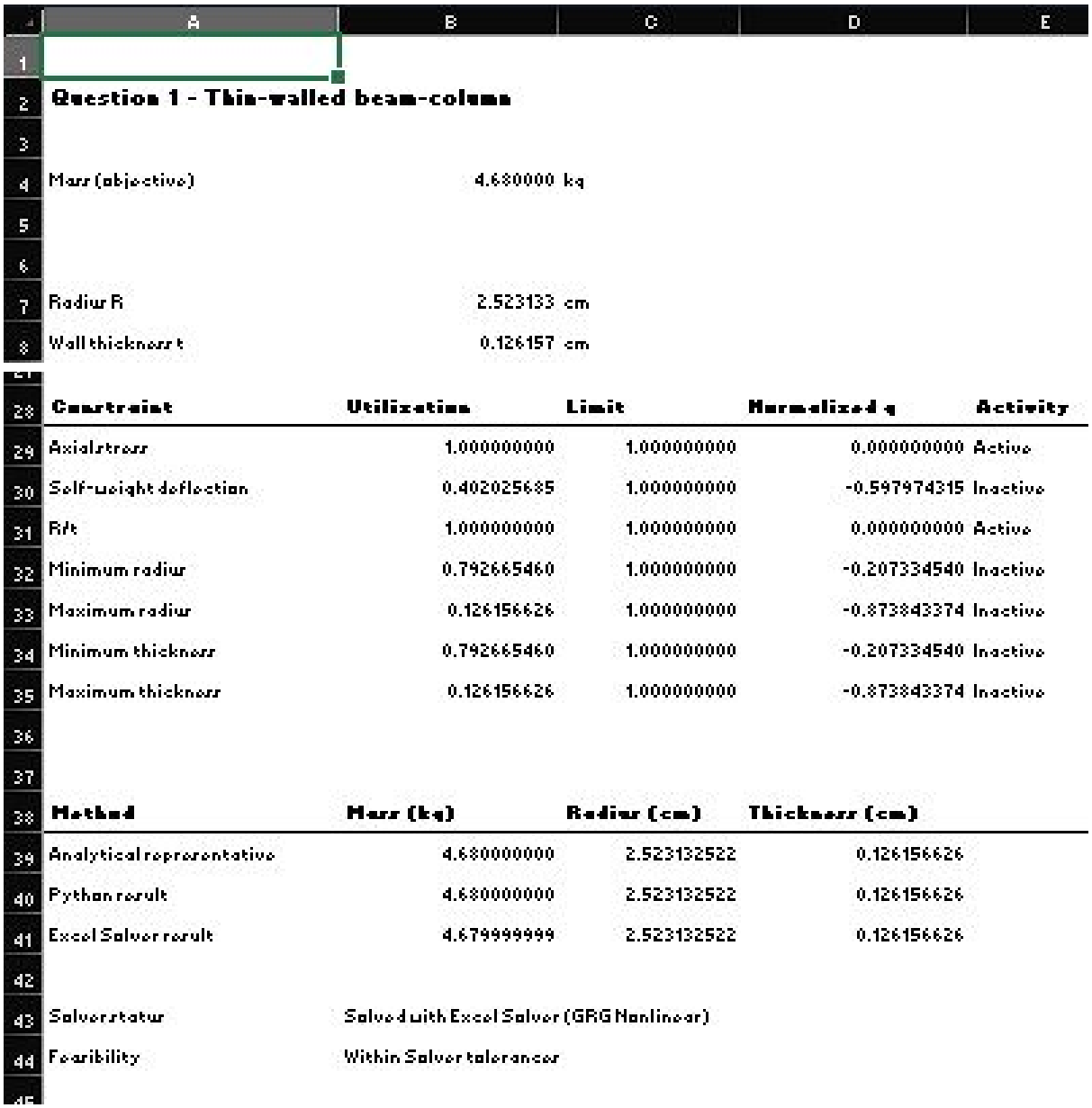
Table 4: Final minimum-mass designs and solver comparison.

Q.	Representative design	Mass (kg)	Active constraints	Δm (kg)
Q1	$R = 2.52313$ cm, $t = 0.126157$ cm	4.68000	Axial stress; R/t	6.8×10^{-10}
Q2	$R_o = 40.0000$ cm, $R_i = 39.8301$ cm	334.448	Bending; outer radius	2.6×10^{-6}
Q3	$H = 50.0000$ cm, $D = 3.42323$ cm	6.60545	Euler buckling; minimum H	5.7×10^{-11}

A Excel worksheet and Solver evidence

The worksheet and Solver Parameters screenshots were captured in Microsoft Excel from a copy of the submitted `CE587_HW1_Solver.xlsx`. The formulas, inputs, saved designs, and Solver models match the submitted workbook. Enlarged crops show the objective, variables, constraint checks, and results. The separate runs in Appendix B confirm convergence from different starting designs.

A.1 Question 1



	A	B	C	D	E
1					
2	**Question 1 - Thin-walled beam-column**				
3					
4	Mass (objective)	4.680000 kg			
5					
6					
7	Radius R	2.523133 cm			
8	Wall thickness t	0.126157 cm			
28	**Constraint**	**Utilization**	**Limit**	**Normalized g**	**Activity**
29	Axial stress	1.000000000	1.000000000	0.000000000	Active
30	Self-weight deflection	0.402025685	1.000000000	-0.597974315	Inactive
31	R/t	1.000000000	1.000000000	0.000000000	Active
32	Minimum radius	0.792665460	1.000000000	-0.207334540	Inactive
33	Maximum radius	0.126156626	1.000000000	-0.873843374	Inactive
34	Minimum thickness	0.792665460	1.000000000	-0.207334540	Inactive
35	Maximum thickness	0.126156626	1.000000000	-0.873843374	Inactive
36					
37					
38	**Method**	**Mass (kg)**	**Radius (cm)**	**Thickness (cm)**	
39	Analytical representative	4.680000000	2.523132522	0.126156626	
40	Python result	4.680000000	2.523132522	0.126156626	
41	Excel Solver result	4.679999999	2.523132522	0.126156626	
42					
43	Solver status	Solved with Excel Solver (GRG Nonlinear)			
44	Feasibility	Within Solver tolerances			

Figure A.1: Q1: actual Excel worksheet at the saved design; objective and variables above, constraint checks and results below.

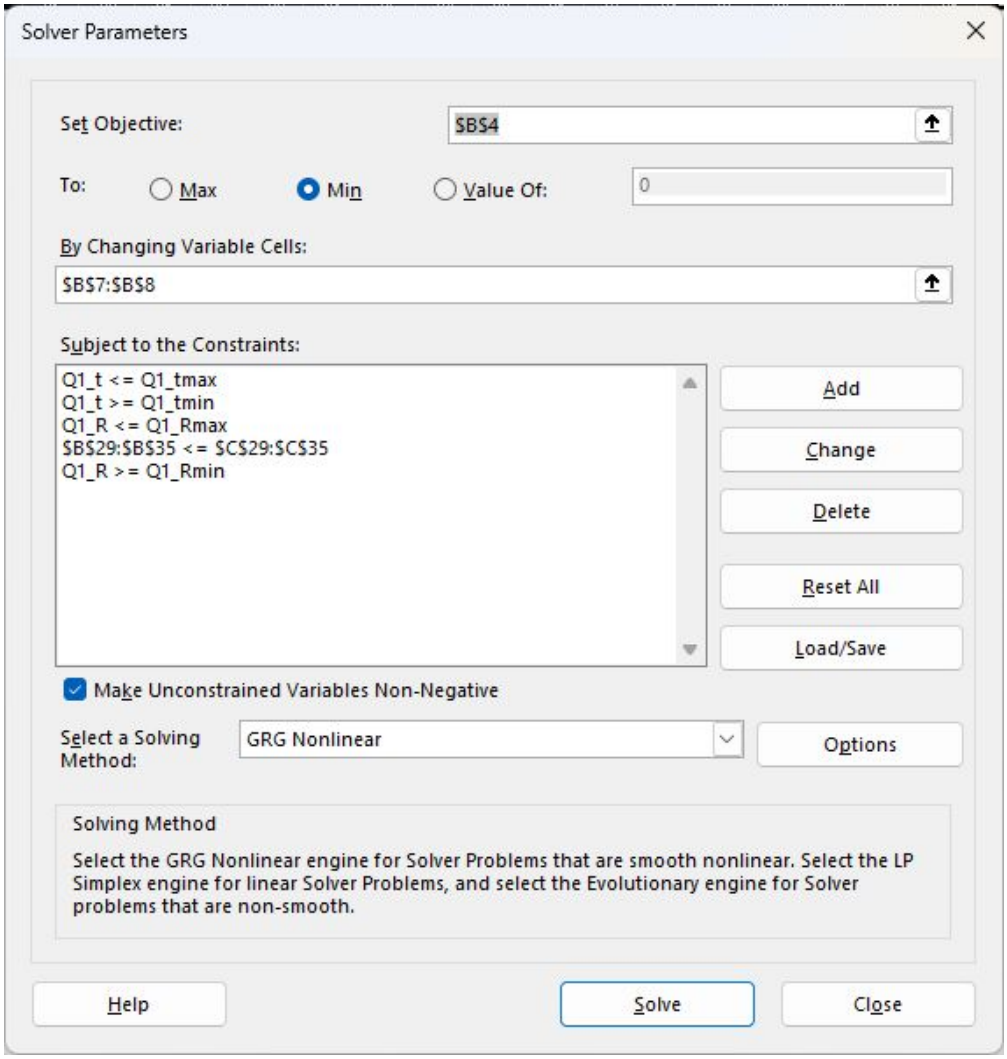


Figure A.2: Q1: actual Excel Solver Parameters dialog showing the loaded model.

A.2 Question 2

	A	B	C	D	E	F	G
1							
2	**Question 2 - Hollow circular cantilever**						
3							
4	Mass (objective)	334.447688 kg					
5							
6							
7	Outer radius Ro	40.000000 cm					
8	Inner radius Ri	39.830120 cm					
24							
25	**Constraint**	**Utilization / lhs**	**Limit**	**Normalized g**	**Activity (tol 1E-6)**		
26	Bending	1.000000	1.000000	0.000000	Active		
27	Shear	0.131440	1.000000	-0.868560	Inactive		
28	Outer radius maximum	1.000000	1.000000	0.000000	Active		
29	Inner radius maximum	0.995753	1.000000	-0.004247	Inactive		
30	Inner radius nonnegative	-0.995753	0.000000	-0.995753	Inactive		
31	Strict geometry (guarded)	-0.004247	0.000000	-0.004247	Inactive		
32							
33							
34	Model: exact annular. cm inputs are converted to m in every response.						
35	Source: official CE587 Homework 1, Fall 2026-2027, page 2.						
36	Ri < Ro is strict mathematically. The 1E-8 m guard is numerical only.						
37							
38	Solver status	Solved with Excel Solver (GRG Nonlinear)					
39	Feasibility	Within Solver tolerances					
40	Solver mass (kg)	334.447688					
41	Solver Ro (cm)	40.000000					
42	Solver Ri (cm)	39.830120					
43							
44	Equivalent smooth Solver constraints (<= 0); Iref = 0.001 m^4						
45	Bending polynomial residual	-2.666450E-09					
46	Shear polynomial residual	-2.947839E-01					
47							

Figure A.3: Q2: actual Excel worksheet at the saved design; objective and variables above, constraint checks and results below.

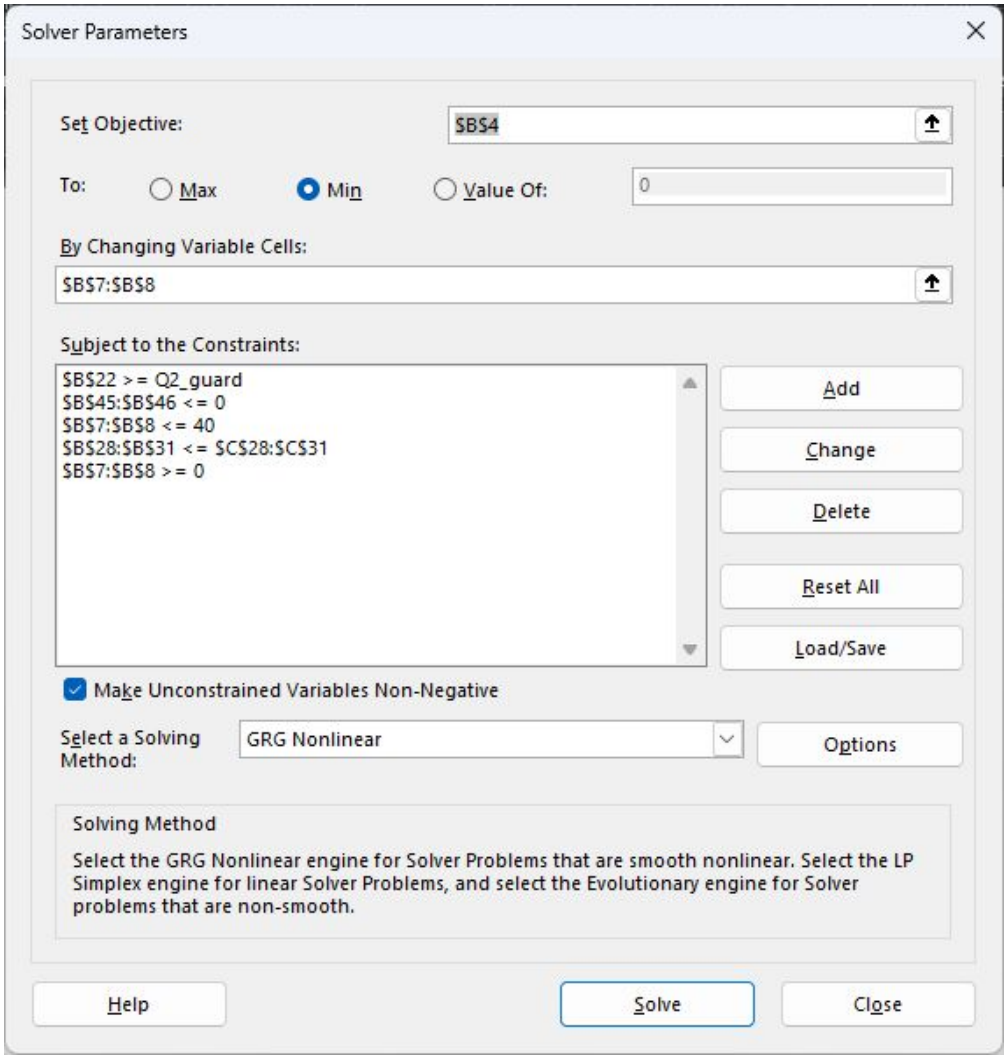


Figure A.4: Q2: actual Excel Solver Parameters dialog showing the loaded model.

A.3 Question 3

	A	B	C	D	E	F	G
1							
2	**Question 3 - Minimum-mass tripod**						
3							
4	Total mass (objective)	6.605448	kg				
5							
6							
7	Height H	50.000000	cm				
8	Strut diameter D	3.423228	cm				
24							
25	**Constraint**	**Utilization**		**Limit**	**Normalized g**	**Activity (tol 1E-6)**	
26	Compression	0.247553		1.000000	-0.752447	Inactive	
27	Euler buckling	1.000000		1.000000	0.000000	Active	
28	Minimum height	1.000000		1.000000	0.000000	Active	
29	Maximum height	0.250000		1.000000	-0.750000	Inactive	
30	Minimum diameter	0.146061		1.000000	-0.853939	Inactive	
31	Maximum diameter	0.342323		1.000000	-0.657677	Inactive	
32							
33							
34	Source: official CE587 Homework 1, page 3, aluminum alloy 2014-T6 table.						
35	Mechanics: N and cm. Density: 2800 kg/m^3 = 0.0028 kg/cm^3.						
36	Euler load uses the prescribed strut length; FS is applied once in row 27.						
37							
38	Solver status	Solved with Excel Solver (GRG Nonlinear)					
39	Feasibility	Within Solver tolerance					
40	Solver mass (kg)	6.605448					
41	Solver H (cm)	50.000000					
42	Solver D (cm)	3.423228					
43							

Figure A.5: Q3: actual Excel worksheet at the saved design; objective and variables above, constraint checks and results below.

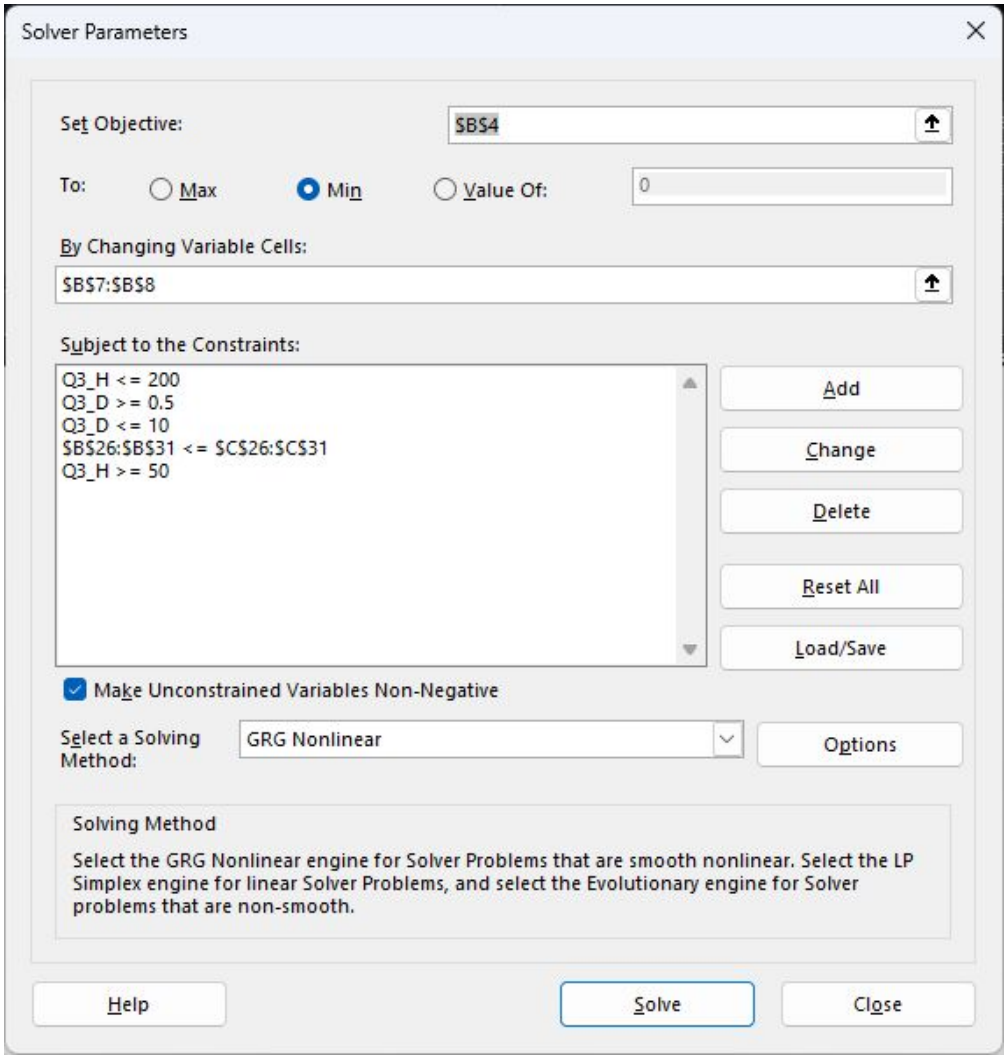


Figure A.6: Q3: actual Excel Solver Parameters dialog showing the loaded model.

B Excel Solver check from different starts

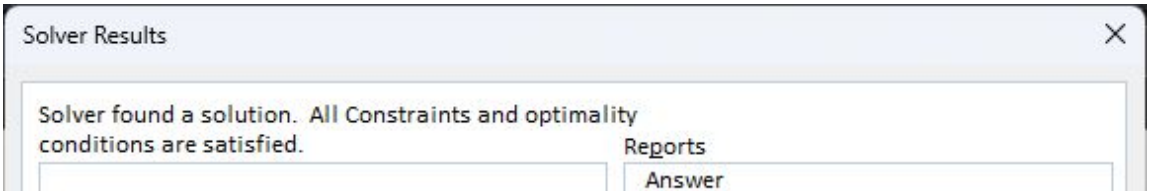
The saved GRG Nonlinear models were run again in Microsoft Excel from the initial designs below. Each run returned "Solver found a solution. All Constraints and optimality conditions are satisfied." The saved design and its mass were reproduced in each case. These numerical checks support the analytical global-optimality arguments in the main solution.

Table 5: Additional Excel Solver runs; all dimensions are in cm.

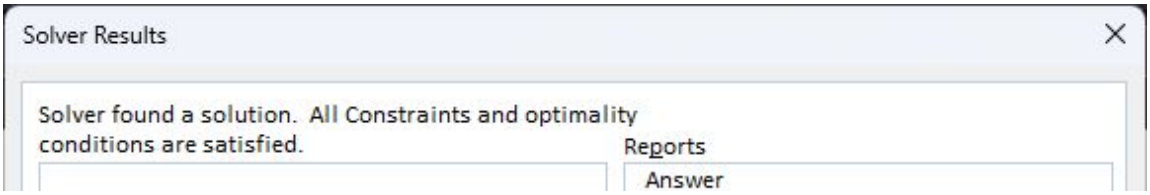
Q.	Starting design	Returned design	Mass (kg)
Q1	$(R, t) = (3, 0.15)$	$(2.52313, 0.126157)$	4.680000
Q2	$(R_o, R_i) = (30, 20)$	$(40.0000, 39.8301)$	334.447688
Q3	$(H, D) = (100, 5)$	$(50.0000, 3.42323)$	6.605448

The maximum positive normalized residual was 1.46×10^{-10} in Q1, within the Solver precision of 10^{-9}. Q2 and Q3 satisfied all normalized inequalities. Substitution of the returned dimensions into the independent response calculations confirmed the masses and constraint checks.

Q1



Q2



Q3

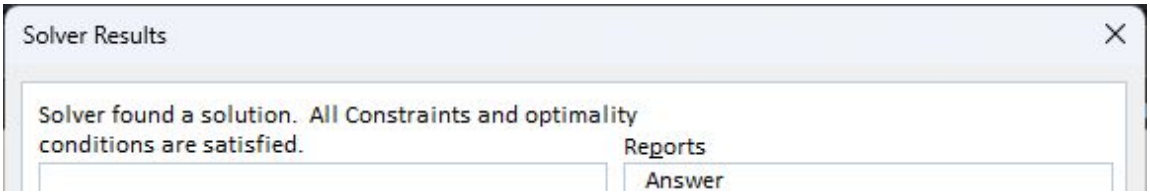


Figure B.1: Actual Excel Solver Results messages from the additional runs. Crops show each success message; full dialog captures are retained with the verification files.